

Shortest path for your bioprocess

from plate to pilot to production

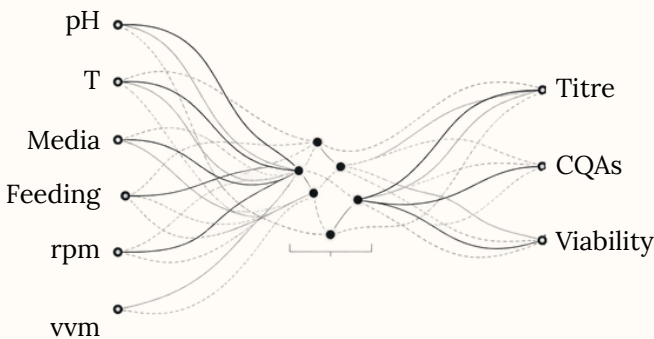


THE PROBLEM

Every bioprocess is a large, partially observed system. You have to make it predictable with surprisingly little data.

From a limited number of experiments, you still need to work out which parameters matter, how they interact, what drives performance and CQAs, and where the process is robust. Most of the design space will never be tested, so a condition that looks good may simply be the best one you happened to sample.

The job is not just to find a process that works. You need enough understanding to know why it works, where it may fail, what remains uncertain, what to test next, and when the process is ready to move forward.



THE SHIFT

Process understanding must compound across the lifecycle & not restart at every stage

Performance, robustness and scale are coupled. Each experiment should refine an evolving understanding of CPP–CQA relationships, uncertainty and scale-dependent behaviour—so every stage begins with a richer, testable understanding of the process.

Old paradigm

understanding resets at every stage.



New paradigm

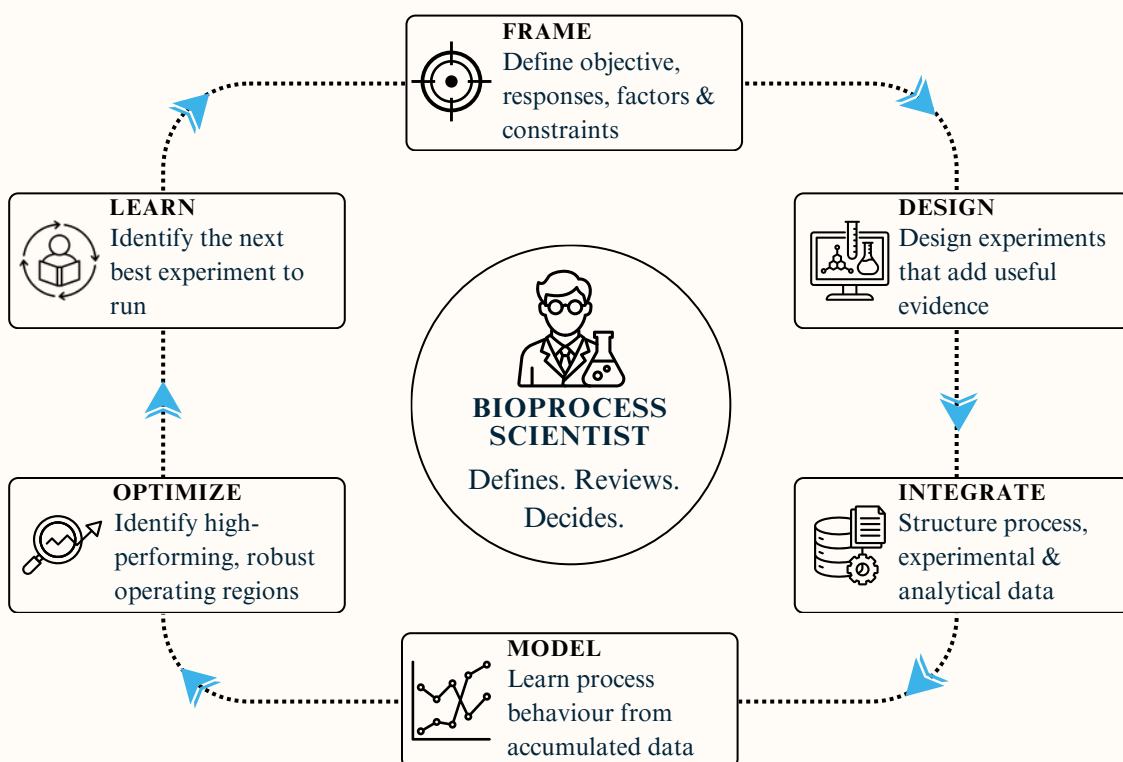
one model accumulates throughout the lifecycle



HOW TRELLIS WORKS

A scientist-led learning loop

Trellis connects experiment design, data, modelling and next-experiment selection in one continuous process-development workflow.



THE NEXT EXPERIMENT IS EVIDENCE-LED

Trellis identifies where another experiment can create the most value.

PROCESS KNOWLEDGE COMPOUNDS

Each round strengthens the same evolving understanding.

THE SCIENTIST STAYS IN CONTROL

Assumptions, choices, uncertainty and recommendations remain visible and reviewable.

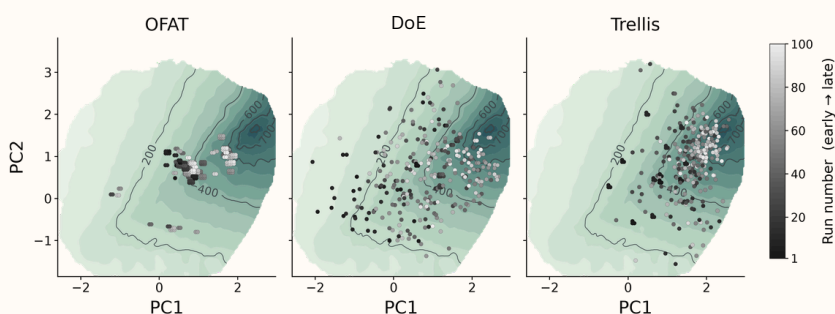
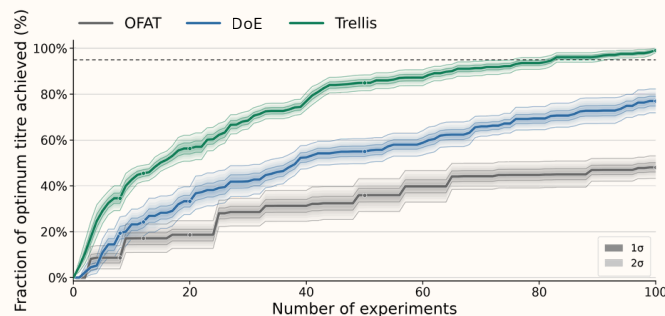
CASE STUDY

Same process. Same objective.

Totally different learning efficiency.

OFAT, DoE and Trellis were benchmarked on the same virtual 12-factor CHO fed-batch process with one objective: maximize titre.

Trellis converges towards the optimum faster.



The difference is where the experiments go.

OFAT searches locally. DoE spreads experiments broadly. Trellis explores uncertain regions, learns from each round and progressively concentrates experiments where better performance is most likely.

START WITH ONE REAL DEVELOPMENT DECISION.



BRING ONE PROCESS

Choose a live process with an important, unresolved question.



DEFINE THE OBJECTIVE

Agree on the objective, available evidence, practical experiments and success measures.



RUN A FOCUSED PILOT

Configure Trellis, generate new evidence and review what the model learns. Decide next steps.

Bring us the development decision you're still uncertain about.



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